

ITT320 INTRO TO COMPUTER SECURITY

CHAPTER 6 : VIRTUAL PRIVATE NETWORK

Zulazeze Sahri
FSKM, UiTM Pahang



Objectives

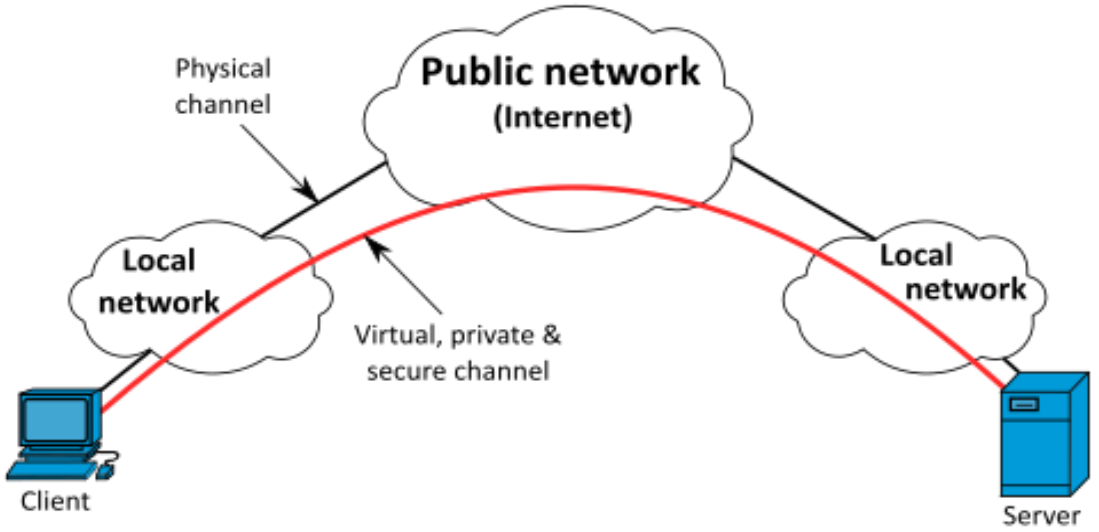
- Use a virtual private network (VPN)
- Use point-to-point tunnelling protocol (PPTP) as an encryption tool for VPNs
- Use layer 2 tunnelling protocol (L2TP) as an encryption tool for VPNs
- Add security and privacy to a communication using IPSec
- Understand and evaluate VPN solutions

Introduction

- Virtual private networks (VPNs) provide users of your network the ability to establish a secure remote connection to the network as if they were connecting locally

- VPNs provide a solution to the administrator's need for security in remote connections while solving the user's need to connect from virtually anywhere

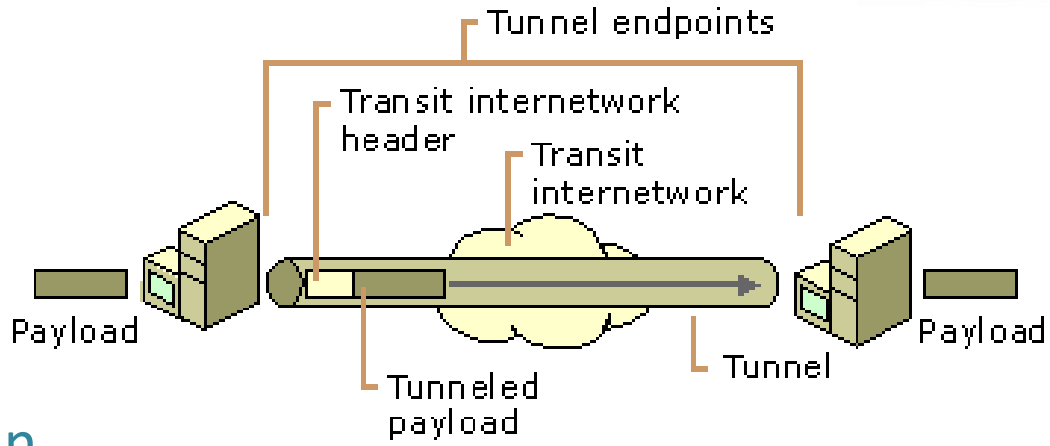
- For telecommuters and traveler who wants to stay connected with intranet (Share file, intranet portal)



Basic VPN Technologies

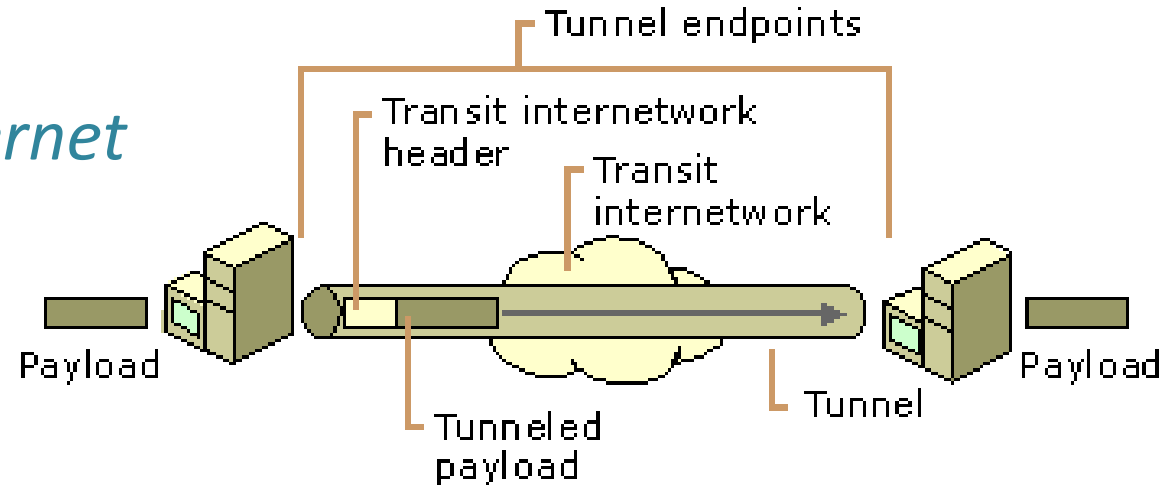
- VPN is based on the idea of Tunneling

- What is VPN Tunneling?
  - Involves establishing and maintaining logical net. Connection (*virtual connection through the internet*)
  - **VPN Protocol** encapsulated the data/packet with an additional header and transmitted over the Internet (VPN Client-Server) through the tunnel (logical conn.).
  - Finally de-encapsulated on the receiving side



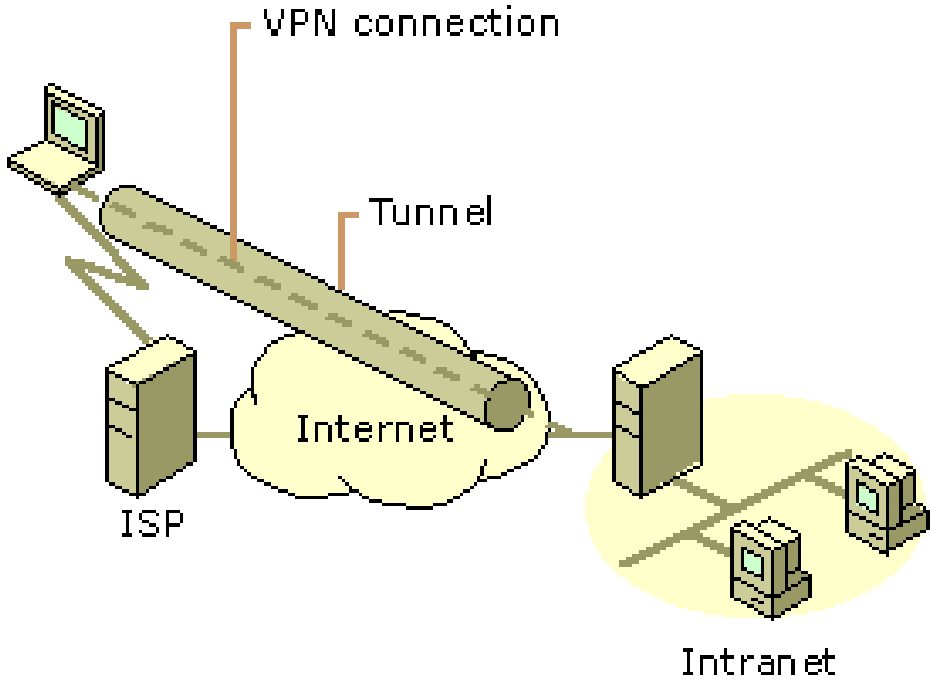
Basic VPN Technologies Cont.

- Does not require additional technologies
- Ability to use public networks like the *Internet* rather than rely on private leased lines.
  - Dial-up and modem bank
  - High-speed (DSL, cable-modem, etc.)



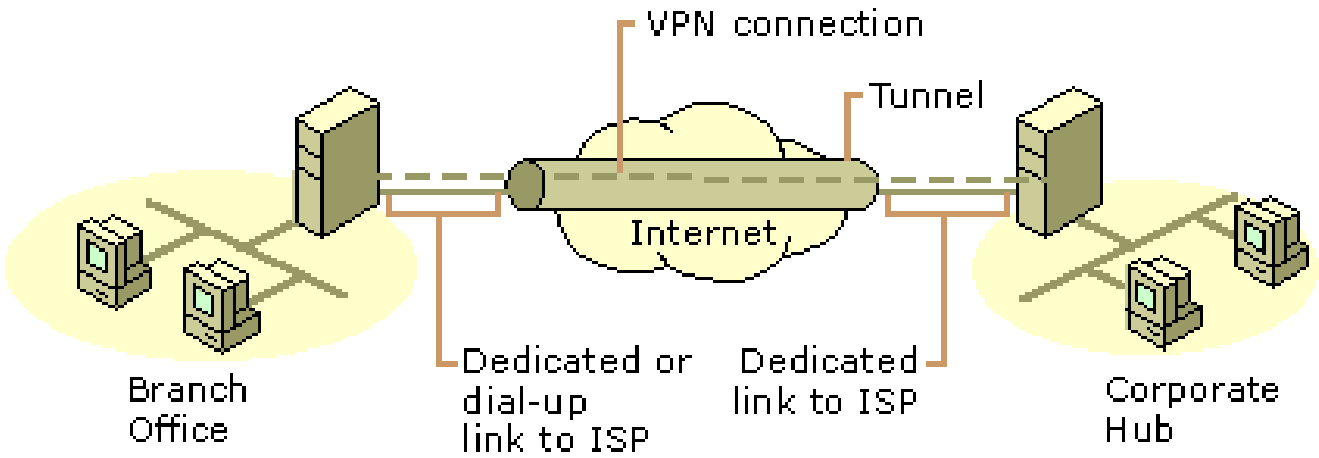
Common Uses of VPNs

Using a VPN connection to connect a remote client to a private intranet



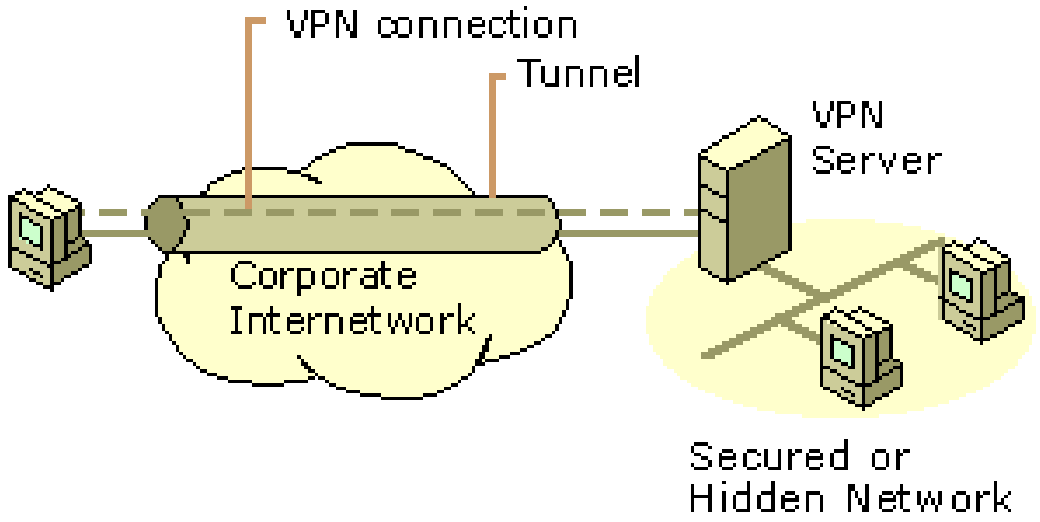
Common Uses of VPNs

- Using a VPN connection to connect two remote sites
- Allows an organization to move away from expensive dedicated data lines



Common Uses of VPNs

Using a VPN connection to connect to a secured or hidden network





VPN Protocols

- Most popular VPN tunnelling protocols as below:

1. Point-to-Point Tunneling Protocol (PPTP) – OSI-L2
2. Layer Two Tunneling Protocol (L2TP) – OSI-L2
3. Internet Protocol Security (IPSec) – OSI-L3

- These protocols focus on **Authentication** and **Encryption** in VPN :

1. Authentication - to correctly establish the identity of people on the network
2. Encryption - potentially sensitive data to be hidden from the general public



VPN Encryption

- MS-CHAP – Microsoft Challenged-Handshake Authentication Protocol
- PAP – Password Authentication
- SPAP – Shiva Password Authentication Protocol
- IPSec – Internet Protocol Security

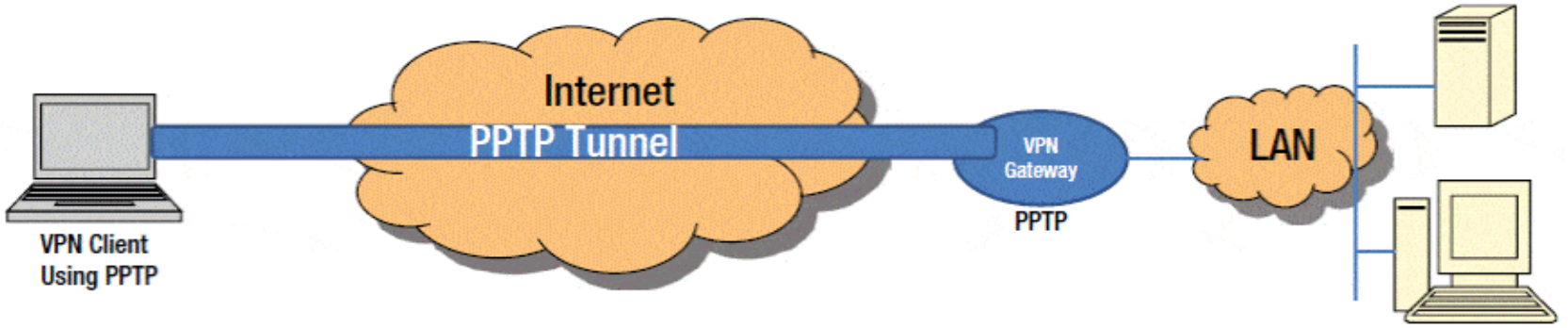


VPN Tunneling Protocols

- PPTP
  - Allows IP, IPX, or NetBEUI traffic to be encrypted, and then encapsulated in an IP header to be sent across Server-Client Tunnel network.
  - Older protocol than L2TP and IPSec
  - Consumes fewer resources than L2TP and IPSec
  - Originally proposed as a standard in 1996
  - Most VPN vendors support it
  - Operates at layer 2 of the OSI model
  - Encrypts packets
    - Microsoft Point-to-Point Encryption (MPPE)

VPN Tunneling Protocols

- PPTP Cont.
- Support 2 types of tunnelling



1. **Voluntary** - Occurs when a workstation or routing server uses tunnelling client software to create a virtual connection to the target tunnel server.

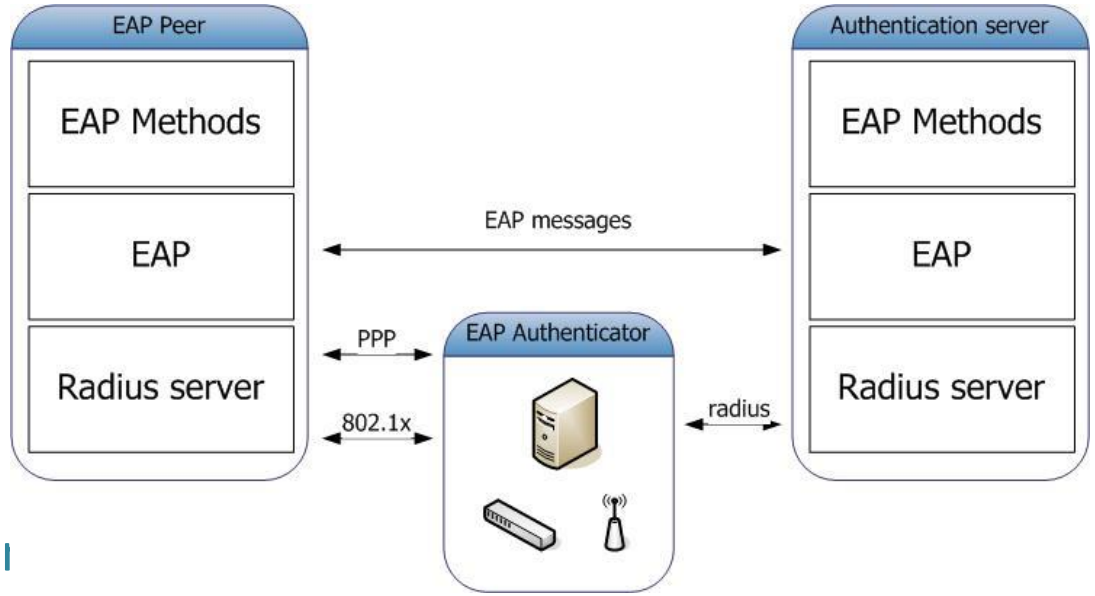
2. **Compulsory** - A VPN-capable dial-up access server configures and creates a compulsory tunnel. With a compulsory tunnel, the user's computer is not a tunnel endpoint. Another device, the dial-up access server, between the user's computer and the tunnel server is the tunnel endpoint and acts as the tunnel client.

VPN Tunneling Protocols

- PPTP Authentication
- Support 2 types of Authentication

1. **Extensible Authentication Protocol** (EAP) – Authentication framework that normally used in wireless networks and Point-to-Point conn.

2. **Challenge Handshake Authentication Protocol (CHAP)** – Authenticates a user or network host to ai authenticating entity (ISP), PERIODICALLY verifies identity of a client by using 3-way-handshake.

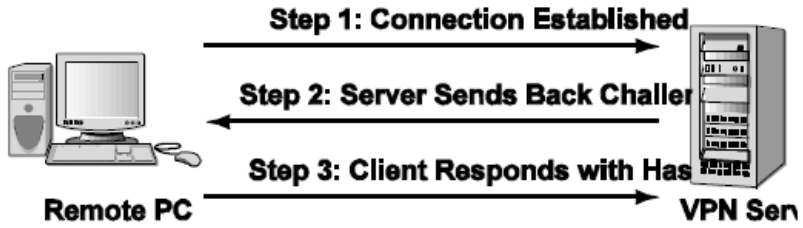


VPN Tunneling Protocols

- PPTP Authentication Cont.

How do CHAP works?

1. After the completion of the link establishment phase, the authenticator sends a "challenge" message to the peer.
2. The peer responds with a value calculated using a one-way hash function on the challenge and the secret combined (client user's pwd).
3. The authenticator checks the response against its own calculation of the expected hash value. If the values match, the authenticator acknowledges the authentication; otherwise it should terminate the connection.
4. At random intervals the authenticator sends a new challenge to the peer and repeats steps 1 through 3.



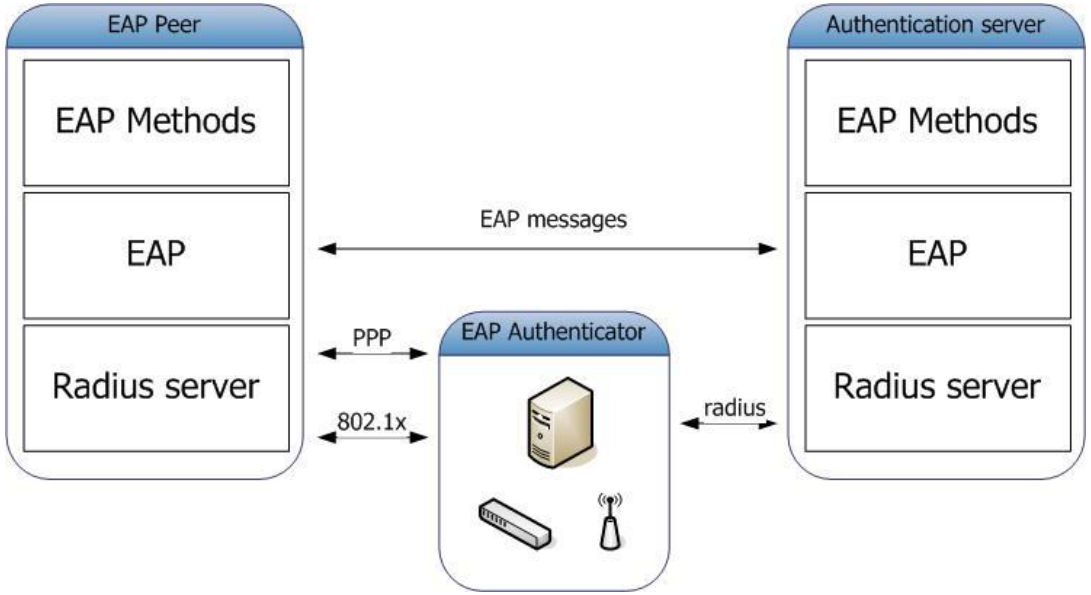


VPN Tunneling Protocols

- L2TP
  - Allows IP, IPX, or NetBEUI traffic to be encrypted, and then sent over any medium that supports point-to-point datagram delivery (Frame Relay, ATM)
  - Enhancement to PPTP
  - Consider less secure than IPSec
  - Not uncommon to be used in conjunction with IPSec

VPN Tunneling Protocols

- L2TP Authentication
  - **EAP**
    - Extensible Authentication Protocol (Wireless / PPP)
    - Rides on top of another protocol such as 802.1x, RADIUS, PPP, etc
    - Allows authenticator to serve as the user authentication carrier between the client and the authenticator server
  - **CHAP**
    - Challenge Handshake Authentication Protocol





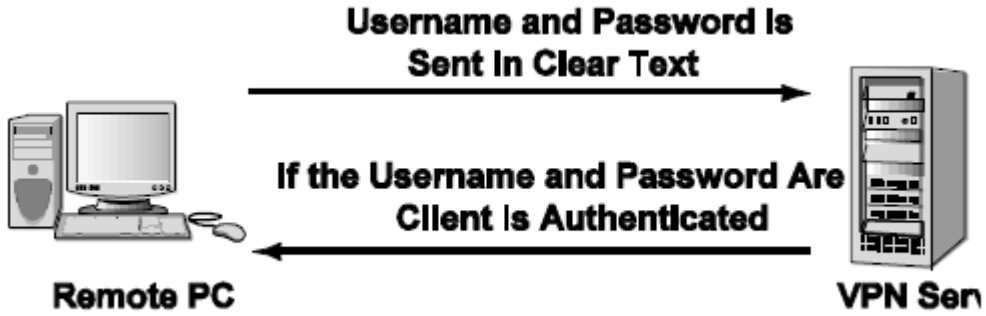
VPN Tunneling Protocols

- L2TP Authentication Cont.
  - **MS-CHAP** – Microsoft-Specific extension to CHAP
  - **MS-CHAP vs CHAP**
    - Response packet formatted for Windows compatibility
    - Does not require the authenticator to store a clear-text or reversibly encrypted password
    - Provides retry and password changing mechanisms
    - Defines a set of reason-for-failure codes

- MS-CHAP-v1 is fundamentally insecure. Tools exist to trivially extract the NT Password hashes from a captured MSCHAP-v1 exchange.
- MS-CHAP-v2 is vulnerable to dictionary attack on the captured challenge response packets.

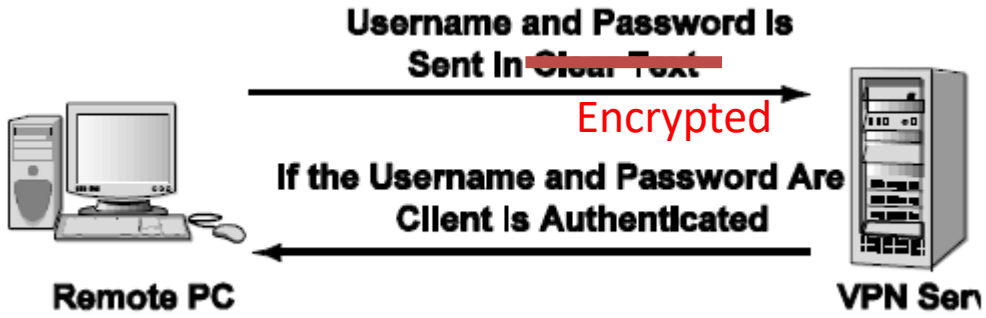
VPN Tunneling Protocols

- L2TP Authentication Cont.
  - PAP
    - “Password Authentication Protocol ”
    - Most basic form of authentication
    - Username and password transmitted in clear text



VPN Tunneling Protocols

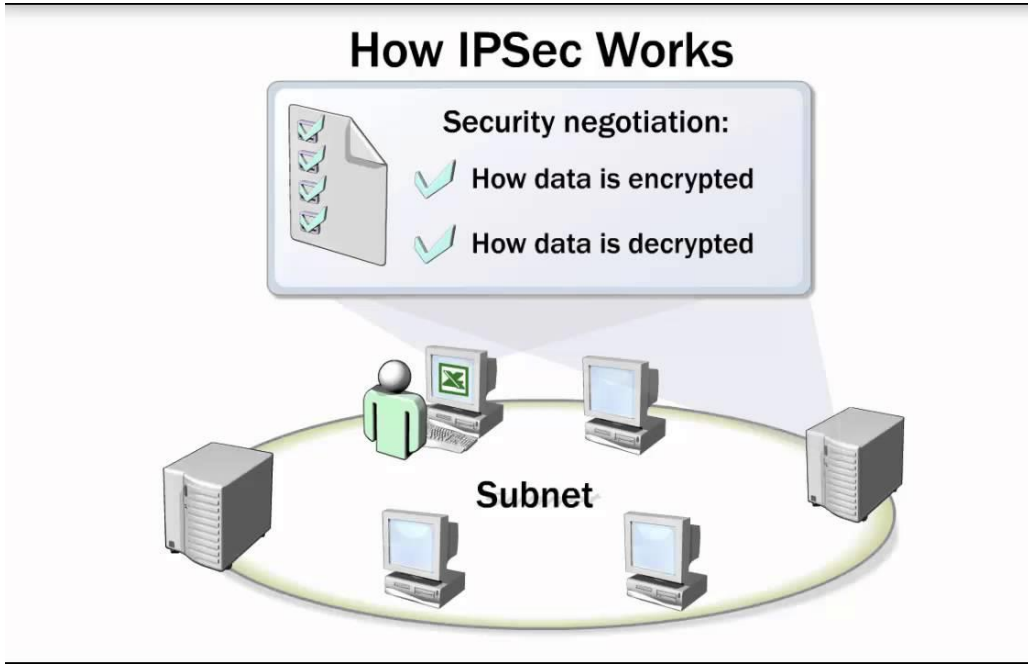
- L2TP Authentication Cont.
  - SPAP
    - “Shiva Password Authentication Protocol ”
    - Proprietary version of PAP
    - Encrypted Username and password
    - Susceptible to “Playback Attack”



IP SECURITY SUITE - IPSec

IPSec – Internet Protocol Security

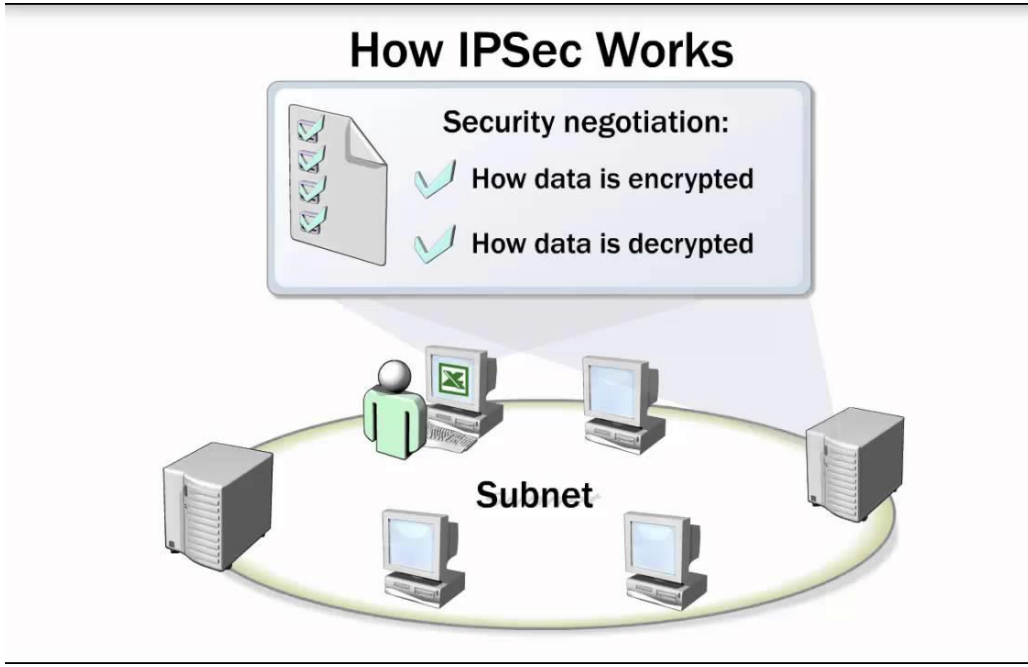
- Used by L2TP for encryption
- Encrypts packet data and header
- Prevents unauthorized retransmission of packets
- Developed by the Internet Engineering Task Force (IETF) - https://www.ietf.org/
- Two encryption modes:
  1. Transport – Encrypt Header only
  2. Tunnel - Encrypt both header and data



IP SECURITY SUITE - IPSec

Other protocols used with IPSec:

1. Authentication Header (AH)
2. Encapsulated Security Payload (ESP)
3. IPComp (IP Payload compression)
4. Internet Key Exchange (IKE)



ODL Homework – find what is the function and how it works?

Implementing VPN Solutions

1. Built-in solutions – using the pre-installed/provided by the operating system
2. Cisco Solutions – various on the shelve vendor based products, e.g, FlexVPN, DMVPN, GET VPN
3. Service Solutions – Managed Security Services by IT Security company
4. Free S/WAN – Example of Linux Free VPN
5. Other solutions





SUMMARY

- VPNs provide a secure way of connecting remote users to the network
- L2TP, PPTP, IPSec are protocols that can be used
- PPTP is based on the PPP
- It is essential to understand the OSI model in relation to how protocols interact and communicate
- There are various VPN solutions that can be implemented
- Solutions built into an operating system may not provide adequate features for large companies